

THE UNIVERSITY OF ZAMBIA
SCHOOL OF HUMANITIES AND SOCIAL SCIENCES
2018/2019 ACADEMIC YEAR FINAL EXAMINATION
DEM 1110: INTRODUCTION TO DEMOGRAPHY

TIME: THREE HOURS

INSTRUCTIONS:

SECTION A: ANSWER ALL QUESTIONS

SECTION B: ANSWER THREE QUESTIONS.

1. ANSWER QUESTION 1 (THIS IS COMPULSORY);
2. ANSWER ONE QUESTION BETWEEN QUESTIONS TWO (2) AND THREE (3); AND
3. ANSWER ONE QUESTION BETWEEN QUESTION FOUR (4) AND FIVE (5)

SECTION A: ANSWER ALL QUESTIONS IN THIS SECTION (40 MARKS)

1. Which concepts of migration – internal and international – do the scenarios below represent?
 - a) Movement of British settlers into the then Northern Rhodesia.
 - b) Zambians who have moved and settled in Botswana.
 - c) People whose original home is a village in Kaputa move out and permanently settle in Wusakile Township in Kitwe.
 - d) People whose original home is Chiwempala Township in Chingola move out and permanently settle in Kabwata Township in Lusaka.
 - e) The departure of educated or professional people from one country from Zambia in search of greener pastures in Europe.
 - f) Movement of Congolese into Zambia because of political persecution and instability in their home country.
 - g) Movement of Nigerians without visas or any documentation into the United States of America in search of economic opportunity.
2. Demonstrate your knowledge of Zambia's population composition by responding to the questions below:
 - a) Which are three largest towns/cities by population size in order of rank?
 - b) Which are the two most densely populated provinces?
 - c) Which are two least densely populated provinces?
 - d) Which are two most densely populated cities?
3. Demonstrate your understanding of population change by responding to the questions below:
 - a) Use the population equation and the data below describe the relative impact of migration and natural increase in Australia between 1954 to 1961.

$$P_t = P_o + B - D + I - O$$

Po (1954)	8,986,530
B	1,544,240
D	600,551
I	1,766,858
E	1,182,104
Pt (1961)	10,514,973

- b) According to the CIA Factbook, Zimbabwe's population has been growing at a rate of 1.56 percent per year. How long will it take for the population to double?
4. Of the examples given below, indicate which are examples of (i) formal demography, (ii) population studies, or (iii) neither:
- Using population projection to estimate the number of school going age population for the Ministry of General Education.
 - A consultancy undertaken by demographers in the Department of Population Studies to review life expectancy tables by NAPSA in their review of the pension scheme.
 - Evaluation of the correctness of age data after the 2010 census by the Central Statistical Office.
 - An analysis of the impact of reduced birth intervals on infant mortality.
 - An analysis of the changing age composition and its impact on consumption patterns.
 - An analysis of the impact of the changing age-sex structure on voting patterns in Zambia.
5. A United Nations expert gives information below at an international conference. For each of the countries given, indicate their stage of aging of their populations.
- That half the population of Malawi is below 16.5 years.
 - That half the population of Japan is below 47.3 years
 - That half the population of Saudi Arabia is below 27.5 years.
6. Demonstrate your knowledge of history of demography by answering the questions below:
- Who between John Graunt and Thomas Malthus was a clergyman?
 - Which two researchers worked on the idea of a logistic curve?
 - What is one major development in demography Edmund Halley is famous for?
 - Other than John Graunt, which other demographer worked on the idea of a life table?
 - When and where was demography for the first time given international recognition as a separate discipline?
 - When and where was the First World Population Conference held?
 - When was the Office of Population Research established?
 - At which university was the Office of Population Research located?
 - Who was the first Director of the Office of Population Research?
 - Mention two regional institutes of population studies funded by the United Nations Population Fund (UNFPA).

- k) When was the Population Council established?
- l) Which prominent American family is associated with the establishment of the Population Council?
- m) When was the Demography Training and Research Program at the University of Zambia established?
- n) Which organization was responsible for the establishment of the demography training and research program at the University of Zambia?
- o) What does the acronym, UNFPA, stand for?

7. Indicate which demographic measure the scenarios below imply.

- a) In India in 1983, for every 1,000 Indians, there were 35 live births.
- b) In Zambia in 2000, for every 1,000 women in reproductive ages (15-49) there 750 children in age group 0-4
- c) There were 88 births per 1,000 women aged 15-49 years in Puerto Rico in 1980.
- d) In Zambia in 2000, the average number of children for women who had stopped having children was seven (7).
- e) In Zambia in 2000, the average number of children in the age group 25-29 was three (3).
- f) In Zambia, the average number of children born to a female, subject to the prevailing age specific rates in 1990 was six (6).
- g) In Bangladesh, the average number of daughters born to a female, subject to the prevailing age-specific fertility rates in 1991 was 1.94.
- h) In Bangladesh, the average number of daughters born to a female, subject to the prevailing age-specific fertility and mortality rates in 1991 was 1.52.
- i) In Zambia in 2014, 49 percent of women were using family planning methods.
- j) For every 1,000 Israelis in 1983, seven (7) of them were reported to have died.
- k) In Alaska in 1990, there about 2 deaths for every 1,000 persons in the age group 15-24.
- l) For every 1,000 live births in 1984, Sweden experienced 6 deaths to infants under age 1.
- m) In the United States in 1983, eight (8) women died for every 100,000 live births due to complications during childbirth.
- n) In Zambia, a child born today is expected to be alive for the next 52.7 years.

8. Demonstrate your understanding of the following statistical concepts:

- a) Inferential statistics
- b) Statistical method
- c) Vital registration system
- d) Parameter
- e) Primary data
- f) Retrospective method of data capture
- g) Categorical data
- h) Proportion
- i) Pattern

SECTION B: ANSWER THREE QUESTIONS: QUESTION ONE (1) IS COMPULSORY; AND CHOOSE ONE QUESTION BETWEEN QUESTIONS TWO(2) AND THREE (3); AND ONE QUESTION BETWEEN QUESTIONS FOUR (4) AND FIVE (5).

1. The Zambia population Based HIV AIDS Impact Assessment (ZAMPHIA) of 2016 collected information on the sexual behavior among people who are HIV positive such as age first at first sex, number of sexual partners and condom use in the last 12 months and whether one had paid for sex. Study table 1 carefully and the questions that follow.

Table 1: Prevalence of HIV among persons age 15-59 years, by sex and sexual behavior characteristics, ZAMPHIA 2016

Characteristic	Males		Females		Total	
	% HIV Positive	Count	% HIV Positive	Count	% HIV Positive	Count
Age at first sexual intercourse						
< 16	9.1	2,179	15.6	2,463	12.1	4,642
16-17	10	1,231	15.1	2,525	13.1	3,756
18-19	10.7	1,332	16.9	2,272	14.2	3,604
≥ 20	12.4	1,787	18.5	1,936	15.1	3,723
Number of sexual partners and concurrent sexual partners (past 12 months)						
0	8.9	1,409	22.6	1,806	15.6	3,215
1	10.4	3,925	14.2	7,091	12.6	11,016
2 or more	12.5	1,121	31.9	266	15.6	1,387
Condom use at last sexual intercourse (past 12 months)						
Used condom	15	1,072	28.2	1,125	20.8	2,197
Did not use condom ✓	9.7	3,967	12.4	6,216	11.2	10,183
No sexual intercourse in the past 12 months						
	8.9	1,409	22.6	1,806	15.6	3,215
Paid for sexual intercourse (past 12 months)						
Yes	15	365	32.2	74	17.4	439
Used condom	15	178	51.1	32	19.7	210
Did not use Condom ✓	14.5	184 ✓	17.2	41 ✓	14.9 ✓	225
No	10.3	4,544	14.7	7,200	12.7	11,744
Total 15-49	8.3	7,358	14.3	10,010	11.4	17,368
Total 15-59	9.3	8,142	14.6	10,973	12	19,115

- a) What is the ratio of those <16 age at first sexual intercourse to the total population of HIV prevalence?
 - i. What does the result mean?

- b) You suspect errors in the age and sex data, what type of error is likely to affect the age data in table 1 (in reporting age at first intercourse)?
 - i. Give methods used in identifying errors associated with age data.
- c) What was the proportion of HIV positive male population who had 2 or more sexual partners and concurrent sexual partners in the past 12 months?
 - i. What does the result mean?
- d) The Ministry of Health have noted with concern that 12% (9.1% for males and 15.6% for females) of population who are HIV positive had their first sexual intercourse when they were below 16 years (<16).
 - i. What type of graph is appropriate to display the distribution of the **age at first sex** of the **total percent of HIV positive** population.
 - ii. Draw the named appropriate graph (part i) showing all the necessary for the chart.
 - iii. Give two (2) consideration you took into account in deciding the type of chart/graph?
 - iv. Give two (2) recommendations you would give the Ministry of Health to address the problems of early initiation of sexual intercourse (those that had first sexual intercourse below age of 16 years (<16)).

2. Demonstrate in a concise and convincing way, with relevant examples, how demographic knowledge is indispensable in the following areas:

- a) Planning for the provision of education
- b) Planning for provision of health facilities,
- c) Planning for provision housing units,

3. Give a concise and convincing argument , with relevant examples, on why population change is responsible for the following problems:

- a) Climate change
- b) Energy shortages
- c) Crime

4. In October 2019, Zambia hosted the Fifth Conference of Ministers responsible for Civil Registration under the theme "Innovative Civil Registration and Vital System: Foundation for Legal Identity Management". With an aim to improve the registration of civil and vital events in Africa. You have been identified as a key Demographer to highlight the problems, solution and the future of Civil and Vital Registration System in Zambia.

- a) Give five (5) general problems affecting the collection of accurate Social and Economic Statistics in Zambia?
- b) What are some of the inherent limitations facing the registration of civil and vital events (statistics) unique to Zambia?
- c) Suggest three (3) clear way you would improve the registration of civil and vital events in Zambia.

5. The Statistical Agency of Zambia (Formerly Central Statistical Office) will be conducting the Census of Population of Housing in 2020. Just like any other data collection method, the Census is prone to errors: